Jian Yan

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Profile: Focused on AI infrastructure and agent systems, with experience in GPU scheduling, training performance diagnostics, real-time conversational infrastructure, and computer-use agent (CUA) environments. Code contributions to TEN Framework, OpenEnv, and OpenAI parameter-golf.

Education

Beijing Jiaotong University | B.Eng. in Communication Engineering
Zhan Tianyou Honors College; Project 985 Platform and 101 Plan

Peking University | Dual Degree in Economics
National School of Development

The Chinese University of Hong Kong | Visiting Student
Department of Electronic Engineering

National Scholarship (2023); municipal second prize, Challenge Cup; municipal award, China International College Students' Innovation Competition.

Project Experience

Huawei Noah's Ark Lab Collaboration | CUA Environments & Harness Ongoing

- **Environment deployment:** Built Docker/KVM desktop environments for OSWorld2; integrated CUA-Gym initialization, desktop actions, and reward scripts into a local execution pipeline, resolving dependency, image, and task-resource compatibility issues.
- **Rollout and evidence capture:** Implemented a DeepSeek-driven observation–action loop, recording per-step screenshots, model outputs, actions, rewards, and logs for reproducibility, failure diagnosis, and trajectory inspection.
- **Evaluation and cross-platform adaptation:** Built setup/reward validation scripts; mapped reset, scoring, and cleanup boundaries across desktop, web, and Android; developed minimal adapters and evaluated Android emulators and extensible environment management.

Agora & OpenAI | Real-Time Conversational AI Infrastructure Jun–Aug 2026

- **Platform tooling:** Contributed in Rust to a unified operations CLI covering deployment, manifest validation, Argo CD diff/sync, releases, and cluster health diagnostics.
- **Test automation:** Built pytest functional, upgrade, and chaos tests for network failures, pod recovery, canary releases, and scaling; reduced critical-path validation from **1–2 person-days to under 30 minutes**.
- **GitOps delivery:** Managed service configuration, environment overlays, and Kubernetes manifests with Argo CD, supporting declarative deployments, change auditing, and rollback across China and overseas clusters.

Agora | LLM Training/Inference Scheduling & Colocation Jun–Aug 2026

- **Resource admission:** Managed a shared GPU pool with Kueue ResourceFlavor and ClusterQueue/LocalQueue across **16 A100 GPUs, 256 vCPUs, and 1 TiB memory**; replaced spreadsheet scheduling with automated queuing and admission.
- **Colocation and validation:** Used cohort borrowing and preemption to reuse idle quota, reclaim capacity for inference scaling, and readmit training jobs; tested quota, StrictFIFO, borrowing, preemption, and recovery after scale-down with pytest.

Agora | Training Observability & Performance Jun–Aug 2026

- **Observability:** Built Prometheus/Grafana/NVIDIA DCGM metrics and dashboards from scratch to compare allocated GPU quota with actual utilization and diagnose training jobs.
- **Bottleneck diagnosis:** Identified insufficient DataLoader parallelism and blocking synchronous checkpoint I/O on rank 0; optimized loading, `/dev/shm`, and disk writes, cutting mean batch time from **1.5 s to 0.5 s** for **3x throughput**.

Open-Source Contributions

TEN Framework | Jev Reasoning-Mode Routing for Voice Agents

github.com/TEN-framework/ten-framework 11,149 Stars

- **Extensions and orchestration:** Developed TypeSafe Jev decision, controller, and model extensions plus an optional voice-assistant graph connecting ASR, routing, LLM, and TTS; selected fast responses or deeper reasoning based on input.
- **Interaction reliability:** Implemented asynchronous cancellation, timeout fallbacks, ASR turn aggregation, and late-event handling; separated routing decisions, reasoning text, and speech output, adding regression tests while preserving the default graph behavior.

Hugging Face OpenEnv | RL Session & Resource Lifecycles

github.com/huggingface/OpenEnv 2,663 Stars

- **Cleanup and ownership:** Fixed provider resource leaks on session creation failure; blocked reallocation after cleanup failures, preserved original exceptions, and made retries and shutdown idempotent without stopping shared resources used by other sessions.
- **Regression coverage:** Added **60 parameterized lifecycle cases** and **4 real Docker fault-injection tests** covering startup, readiness, child-session construction, cleanup failures, and recovery ordering.
- **FastMCP 4 migration:** Adapted session-state isolation, MCP wire schemas, and harness sampling guards while preserving Gym API boundaries; core suite: **834 passed / 1 skipped**.

OpenAI parameter-golf | Activation Experiments under a Fixed Budget

github.com/openai/parameter-golf 5,198 Stars

- Replaced MLP ReLU^2 with $\text{LeakyReLU}(0.5)^2$ to retain gradients for negative inputs; submitted baseline improvement code and experimental results.
- **Local experiments:** On two RTX 5090 GPUs with a 600 s budget, bpb decreased from **1.3822 to 1.2947**; the two runs completed 1,000 and 1,947 steps, respectively. Results reported for these local conditions.

Research Experience

National Key R&D Program | Space Science Data Detection IEEE DLCV 2025 | First Author / Oral

- Contributed to the National Space Science Center, Chinese Academy of Sciences project on intelligent management and analysis of space-science big data, studying ionogram reconstruction and detection. Compared five models on **4,311 Iono4311 ionograms**; DETR achieved **54.2% accuracy / 9.2 FPS**, exceeding YOLOv3 by **10.0 percentage points** in accuracy and approximately **26%** in speed. Released paper code and training methods.

Medical Imaging | Deep Learning for Phase Unwrapping Algorithm Research / Implementation

- Collaborated with research teams associated with CUHK and MIT on neural phase-unwrapping algorithms, training validation, and engineering implementation; improved accuracy by **18%**. Advanced a first-author manuscript submission and a Chinese invention patent application.

Engineering Practice & Technical Communication

- **Medical imaging deployment:** Contributed to C++ algorithm refactoring and on-device GPU acceleration at Beijierui (倍捷锐). The project was exhibited at HICOOL, and imaging equipment was used at Tsinghua-affiliated medical institutions.
- **Edge inference and knowledge sharing:** Deployed local inference on Raspberry Pi with llama.cpp; investigated TensorRT and tiered OSS storage and presented findings to the team. Publish technical notes under "Jian Yan's Learning Notes"; an agent-focused article received over **1,000 views**.

Technical Skills

Systems & Infrastructure: Python, Rust, C++, Linux, Docker/KVM, Kubernetes, Kueue, Argo CD.

Training & Agents: PyTorch, DDP/NCCL, Prometheus/Grafana, pytest, MCP, rollout/evaluation harnesses.